DECENT SPACES OF FINITE TYPE OVER A JACOBSON BASE ARE JACOBSON

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Scheme case over a Jacobson base. Morphisms, lemma Jacobson-etale (cross-chapter, named). Etale morphisms of schemes over a Jacobson base are Jacobson.
- Jacobson is etale-local. spaces-properties lemma Jacobson-local (cross-chapter, named). Etale-local descent of the Jacobson property.

Source reference. Original lemma 0BA5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5511–5511. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03K5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2865–2865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06EK: focused-seeds/geometry/S9-decent-decent-jacobson/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02I7: focused-seeds/geometry/S9-decent-decent-jacobson/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Then X is Jacobson if and only if $|X|$ is Jacobson.*

Proof. Assume X is Jacobson and that $T \subset |X|$ is a closed subset. By Morphisms of Spaces, Lemma 06EK we see that $T \cap X_{\text{ft-pts}}$ is dense in T. By Lemma 0BA5 we see that $X_{\text{ft-pts}}$ are the closed points of $|X|$. Thus $|X|$ is indeed Jacobson.

Assume $|X|$ is Jacobson. Let $f : U \to X$ be an étale morphism with U an affine scheme. We have to show that U is Jacobson. If $x \in |X|$ is closed, then the fibre

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$F = f^{-1}(\{x\})$ is a finite (by definition of decent) closed (by construction of the topology on $|X|$) subset of U. Since there are no specializations between points of F (Lemma 03K5) we conclude that every point of F is closed in U. If U is not Jacobson, then there exists a non-closed point $u \in U$ such that $\{u\}$ is locally closed (Topology, Lemma 02I7). We will show that $f(u) \in |X|$ is closed; by the above u is closed in U which is a contradiction and finishes the proof. To prove this we may replace U by an affine open neighbourhood of u. Thus we may assume that $\{u\}$ is closed in U. Let $R = U \times_X U$ with projections $s, t : R \to U$. Then $s^{-1}(\{u\}) = \{r_1, \ldots, r_m\}$ is finite (by definition of decent spaces). After replacing U by a smaller affine open neighbourhood of u we may assume that $t(r_j) = u$ for $j = 1, \ldots, m$. It follows that $\{u\}$ is an R-invariant closed subset of U. Hence $\{f(u)\}$ is a locally closed subset of X as it is closed in the open $|f|(|U|)$ of $|X|$. Since $|X|$ is Jacobson we conclude that $f(u)$ is closed in $|X|$ as desired. $\square$